

HUAILAI, China

WMO: 544050

Lat: 40.4174N

Lon: 115.5071E

Elev: 1873

StdP: 13.73

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	7.9	-22.5	1.7	14.6	-18.1	2.2	16.6	26.9	12.8	24.6	15.1	10.7	290	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	92.3	67.1	89.6	67.2	87.2	67.2	75.2	84.7	73.7	82.6	72.1	80.8	7.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	128.7	80.3	70.9	122.2	79.4	69.3	115.6	78.1	40.3	85.3	38.8	82.6	37.3	80.6	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.0	16.6	-0.2	97.5	4.0	3.2	-3.0	99.7	-5.4	101.6	-7.6	103.4	-10.5	105.7
			-2.0	77.5	3.8	1.6	-4.7	78.7	-6.9	79.6	-9.0	80.5	-11.7	81.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	20.6	27.5	40.6	54.9	66.0	73.6	77.3	74.6	65.5	52.2	36.4	23.9	
	DBStd	20.92	6.64	7.23	8.17	7.81	6.39	5.15	4.13	4.04	5.72	6.72	7.86	6.54	
	HDD50	3163	910	630	310	39	1	0	0	0	0	54	410	809	
	HDD65	6157	1375	1050	758	312	63	4	0	1	62	400	858	1274	
	CDD50	3605	0	0	17	186	498	708	846	763	464	121	2	0	
	CDD65	1126	0	0	0	9	95	263	382	299	76	2	0	0	
	CDH74	10017	0	0	5	153	1023	2404	3493	2334	583	22	0	0	
	CDH80	3497	0	0	0	31	329	929	1349	731	126	2	0	0	
Wind	WSAvg	6.4	8.1	7.3	7.4	7.2	7.0	5.5	5.0	4.6	4.7	5.5	6.4	7.7	
Precipitation	PrecAvg	15.2	0.1	0.1	0.3	0.6	1.0	2.2	4.4	4.0	1.5	0.7	0.2	0.0	
	PrecMax	21.4	0.5	1.0	0.9	3.0	2.4	5.3	10.4	7.9	3.4	2.0	0.7	0.2	
	PrecMin	7.6	0.0	0.0	0.0	0.0	0.1	0.5	1.7	0.8	0.2	0.0	0.0	0.0	
	PrecStd	3.3	0.1	0.2	0.2	0.6	0.6	1.1	2.2	1.9	0.9	0.6	0.2	0.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	52.2	71.2	83.7	92.3	95.9	96.7	92.6	88.4	76.5	61.4	44.5	
		MCWB	31.1	36.8	48.6	56.3	60.8	63.7	69.0	70.6	64.7	57.3	44.7	32.7	
	2%	DB	37.9	47.6	64.2	78.1	87.4	91.7	92.7	89.1	83.4	71.6	56.4	40.2	
		MCWB	28.2	34.5	44.5	53.2	59.3	64.5	69.7	69.9	63.8	54.1	42.4	30.3	
	5%	DB	34.6	43.8	59.3	73.8	83.7	88.7	89.7	86.5	80.1	68.0	52.0	37.0	
		MCWB	26.0	32.4	42.4	51.8	58.2	64.3	69.9	69.8	62.3	52.3	40.2	28.2	
	10%	DB	31.4	40.1	54.7	69.5	79.9	85.5	87.1	84.2	77.1	64.2	48.3	34.1	
		MCWB	23.9	30.3	39.7	49.8	56.9	64.0	70.2	69.2	61.2	51.0	38.7	26.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.6	39.0	50.4	60.8	66.4	72.1	77.4	77.1	70.9	61.0	47.1	33.6
			MCDB	41.5	47.7	65.6	72.9	80.8	84.3	87.0	86.4	81.0	70.3	55.4	42.3
2%		WB	28.6	35.5	46.3	57.1	63.8	70.3	75.6	75.1	68.0	58.1	44.5	30.7	
		MCDB	37.2	46.4	62.0	72.0	78.7	82.3	85.6	83.6	76.6	66.7	53.1	39.2	
5%		WB	26.3	33.0	43.2	54.4	61.9	68.7	74.4	73.5	65.9	55.6	41.8	28.8	
		MCDB	34.0	43.1	57.4	69.0	75.5	80.2	84.0	81.5	75.2	63.6	50.7	36.2	
10%		WB	24.1	30.6	40.6	51.9	60.0	67.2	73.0	71.8	63.9	53.1	39.2	26.8	
		MCDB	31.0	39.5	53.4	65.8	73.7	78.7	82.1	79.6	73.5	61.1	47.6	33.6	
Mean Daily Temperature Range		5% DB	MDBR	16.0	18.3	20.1	21.6	21.9	19.9	17.1	17.8	19.8	19.0	16.3	14.9
			MCDBR	19.4	23.2	26.7	28.6	28.9	26.1	22.6	21.6	24.3	24.0	21.4	18.4
	MCWBR		13.2	14.4	14.2	12.4	10.6	8.0	6.9	7.4	9.0	11.0	12.1	12.0	
	5% WB	MCDBR	19.0	22.1	24.7	24.0	22.1	19.1	16.7	16.6	19.0	19.9	19.7	17.9	
		MCWBR	13.0	14.0	13.6	11.0	9.0	6.9	6.6	6.8	8.2	10.0	11.7	11.9	
Clear-Sky Solar Irradiance	taub	0.351	0.438	0.464	0.535	0.592	0.714	0.690	0.668	0.579	0.506	0.391	0.359		
	taud	2.139	1.931	1.909	1.809	1.708	1.540	1.624	1.645	1.766	1.862	2.103	2.162		
	Ebn at Noon	246	235	246	238	227	200	204	203	211	210	227	232		
	Edh at Noon	34	49	56	66	74	88	80	76	63	52	35	31		
All-Sky Solar Radiation	RadAvg	835	1104	1502	1786	1977	1789	1674	1623	1416	1127	846	732		
	RadStd	46	72	114	111	144	154	102	100	76	85	94	49		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page